

PH113 / Assignment - Module 1: Physical Optics

(Dated: January 25, 2024)

Here, we present some of the *benchmark* question, mostly numerical, from Module I to be discussed in the tutorials. It is expected that the students will solve these questions on their own, or in groups before the tutorial(s). At the same time, questions similar to the ones contained herein are likely to be asked in your exams. However, note that (i) same questions should NOT be expected in the exams, (ii) numerical questions beyond these could also be asked in the exams. In other words, this document is NOT a question bank of any kind. Therefore, the students should aim to understand the underlying concepts well.

1. Malus Law.

- (a) Consider a geometry of sequence of two polarizers rotated at a relative angle of 45° from each other. What would be resulting amplitude and intensity if the incident unpolarized (light) beam had the intensity I_0 ?

Solution. The behaviour of light passing through a polarizer is given by Malus law. However, some care should be taken when the incident light is unpolarized. When unpolarized light passes through a polarizer, its intensity is given by:

$$\begin{aligned} I &= I_0 \langle \cos^2 \theta \rangle \\ &= I_0 \cdot \frac{1}{2\pi} \cdot \int_0^{2\pi} \cos^2 \theta d\theta \\ &= I_0/2, \end{aligned}$$

where, $\langle \dots \rangle$ denotes the average value. Therefore, after the 1st polarizer, intensity will be $I_1 = I_0/2$.

When a *polarized* light passes through a polarizer, its intensity is given by Malus law, which states that the resulting intensity is: $I = I_0 \cos^2 \theta$, where θ is the relative angle between the polarization of the incident light and the polarizer. Therefore, after the second polarizer $I_2 = I_1 \cdot \cos^2(\pi/4) = (I_0/2) \cdot \cos^2(\pi/4)$.

The amplitudes can, in general, be obtained by $A = \sqrt{I}$. Note, however, that the amplitude after the first polarizer **cannot** be obtained by averaging $\cos \theta$ over any

period – **this only works for intensities!**.

- (b) Similar to above, consider a sequence of three polarizers at relative angles of 20° to each other. Also, the first polarizer is aligned along the y -axis. Compute the intensities I_i ($i = 1, 2, 3$) after each polarizer if the intensity of the y -polarized source is given to be $I_0 = 1000 \text{ W/m}^2$.

Solution. Since the initial beam is polarized along the y -axis, it doesn't affect the intensity of a y -polarized light. All subsequent polarizers have the same effect: $I_i = I_0 \cdot \cos^{2(i-1)}(20^\circ)$, where $i = 1, 2, \dots n$.

- (c) What happens to the final intensity if the second polarizer is removed?

Solution. Intensity should decrease even if a polarizer is removed since we'll now work with $\cos^2(40^\circ)$.

- (d) [!] It is rather straight-forward to generalize the above situation to n polarizers separated by angle θ (fixed!). In principle, it would be interesting to find out when the resulting intensity will be $\lesssim 0.10I_0$.

Solution. For θ angles between the polarizers and xI_0 to be the resultant intensity:

$$\begin{aligned} I_0 \cos^{2n} \theta &= xI_0 \\ \cos \theta &= x^{1/2n} \\ \implies n &= \frac{1}{2} \frac{\log(\cos(\theta))}{\log(x)} \end{aligned}$$

*Comment: **At-home exercise:** Find Polaroid sunglasses and rotate one while holding the other still and look at different surfaces and objects. Explain your observations. What is the difference in angle from when you see a maximum intensity to when you see a minimum intensity? Find a reflective glass surface and do the same. At what angle does the glass need to be oriented to give minimum glare?*

2. Brewster angle

On a calm surface of a pond or a lake (interface between air and water), for which position of the sun would the reflected light from the interface be completely linearly polarized. The refractive indexes for air and water should be taken as $n_{\text{air}} = 1$ and $n_{\text{water}} = 1.33$.

| *Comment: This situation is, in fact, realized at the beaches or lakes.*

Solution. The Brewster's law for producing linearly polarized reflected light between a rarer and a denser medium is:

$$\theta_B = \tan^{-1} \left(\frac{n_2}{n_1} \right). \quad (1)$$

For interface between water and air, $\theta_B = 53^\circ$ using $n_2 = n_{\text{water}}$ and $n_1 = n_{\text{air}}$. Therefore, the sun should be 37° above the ground (horizontal).

3. A plate of flint glass (refractive index 1.67) is immersed in water (refractive index 1.33). Calculate the Brewster angle for internal as well as external reflection at an interface.

Solution. The external surface is formed between air/water, and the internal surface is formed between water/glass. Other part(s) to the solution is rather straight forward and clues can be drawn from prev question.

4. [!] At a particular beach on a particular day near sundown the horizontal component of the electric field vector is 2.3 times the vertical component. A standing sunbather puts on polarizing sunglasses; the glasses suppress the horizontal component. (a) What fraction of light intensity received before the glasses were put on now reaches the eyes? (b) The sunbather, still wearing the glasses lies on his side. What fraction of the light energy received before the glasses were put on reaches the eyes now?

Solution. Intensity is proportional to the electric field squared, so the original intensity reaching the eye is I_0 , with components $I_h = (2.3)^2 I_v$, and then $I_0 = I_h + I_v = 6.3 I_v$ or $I_v = 0.16 I_0$. Similarly, $I_h = (2.3)^2 I_v = 0.84 I_0$. (a) When the sun-bather is standing only the vertical component passes, while (b) when the sun-bather is lying down only the horizontal component passes

5. Double refraction.

A thin quartz crystal of thickness t has been cut so that the optical axis is $\parallel$ to the surface of the crystal. For sodium light ($\lambda = 589 \text{ nm}$), the refractive index of the crystal is 1.55 for e-ray and 1.54 for o-ray. If two beams of light polarized $\parallel$ and $\perp$ O.A., respectively, start through the crystal in-phase, how thick must the crystal be if the emerging rays have a phase difference of $\pi/2$.

Solution. Time taken by the e-ray to cross the thickness t : $t_e = t/v_e = t/(c/n_e) = n_e t/c$ with $n_e = 1.55$.

Similarly, time taken by the o-ray: $t_o = n_o t/c$.

Time spent by the e-ray inside the crystal is longer. In the time difference $t_e - t_o$, the o-ray travels in vacuum/air.

Therefore, the path difference between the two rays when they emerge from the other side:

$$c(t_e - t_o) = c\left(\frac{n_e t}{c} - \frac{n_o t}{c}\right) = (n_e - n_o)t$$

Corresponding phase difference:

$$\phi = \frac{2\pi}{\lambda}(n_e - n_o)t.$$

Equating this (p.d) to $\pi/2$, we get:

$$t = \frac{\lambda}{4(n_e - n_o)} = \frac{589 \text{ nm}}{4 \times 0.01} = 14.7 \mu\text{m}.$$

| *Comment: What is the path difference here?*

6. Calculate the thickness of a doubly refracting crystal required to introduce a path difference of $\lambda/2$ between the e-ray and o-ray when $\lambda = 60000 \text{ \AA}$, $\mu_o = 1.65$ and $\mu_e = 1.54$.

Solution. Thickness t is given by:

$$\begin{aligned} t &= \frac{\lambda}{2(\mu_o - \mu_e)} \\ &= 2.727 \times 10^{-6} \text{ m} = 2.727 \mu\text{m}. \end{aligned}$$

7. Plane waves and their superposition

- (a) Plot $y = y_0 \cos(\omega t - kx + \phi)$ for (i) $\omega = 1 \text{ s}^{-1}$, (ii) $\omega = 0.5 \text{ s}^{-1}$ and $\phi = 0, \pm\pi/4, \pm\pi/2, \pm\pi, \pm2\pi$. What are the units of ϕ , ω and k ?

Solution. The idea is to explain to the students that a time snapshot of the wave looks the same as looking at the wave amplitude at a single point (fixed x) in space as a

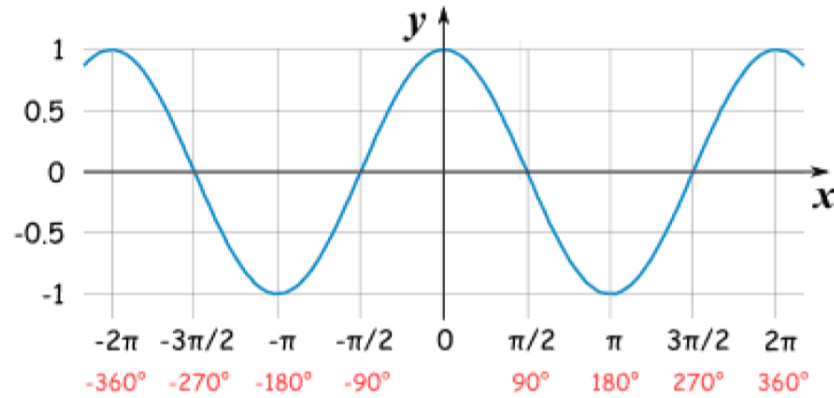


FIG. 1. Plane wave $y(x, t) = A \cos(\omega t - kx)$.

function of t . This is the essence of a *travelling wave*.

One could discuss how to visualize the phase difference (the wave moves left/right), etc. and the dispersion relation $\omega = ck$ which, in fact, a central relation.

To put it in words (borrowed from the below stackexchange link), It is an expression describing a travelling wave. More precisely, it says

$$y(x, t) = A \cos(\omega t - kx + \phi)$$

where x is the coordinate in space (location along a line in the direction where the wave is moving), t is time, $k = 2\pi/\lambda$ where λ is the wave length, and the Greek letter omega $\omega = 2\pi f$ is the angular frequency (frequency in hertzes multiplied by $2\pi \approx 6.28$).

The function $\cos$ is the cosine, a function that oscillates between -1 and $+1$. The graph of the function is above. It may be imagined *e.g.* as the height of a valve on a wheel of a car as a function of time — or the location of the end of the spring that oscillates. The oscillations that depend on time through the cosine are known as the harmonic oscillations.

The number A is just a coefficient, the maximum amplitude, that dictates the strength of the oscillations. Because the cosine is between -1 and $+1$, $A \cos(\dots)$ is between $-A$ and $+A$.

The argument of the cosine isn't just ωt but $(\omega t - kx)$. If we choose some appropriate units, it is the difference between the position and time. It means that the quantity $y(x, t)$ depends on the position in space x in the same way as it depends on time — **so even at one moment, it looks like the cosine wave in space.**

There is a difference there. If we increase ωt and kx by the same constant α , the terms α cancel in the total phase $(\omega t - kx)$, so the value of $y(x, t)$ is the same. It means that waiting for some time has the same effect as looking to the right: the wave is moving to the right. It is moving by the speed $c = \omega/k$ which can be extracted as the slope of the line given by the equation $kx - \omega t = \text{constant}$.

The periodicity of the cosine is 2π . So if the argument of the cosine is changed by 2π (radians), we get the same value of the cosine. It means that $y(x, t)$ is a periodic (repeating) function of time as well as the space. The periodicity in time is $\Delta t = 2\pi/\omega$ because $\omega \Delta t = 2\pi$ while the periodicity in the spatial coordinate λ is the so-called wavelength (the distance between two maxima, for example), $\Delta x = 2\pi/k$, for an analogous reason.

| *Comment: This should be useful.*

- (b) Consider superposition of two plane waves of equal amplitude and in-phase ($\phi = 0$), polarized along $\hat{x}$ and $\hat{y}$, respectively. What would be resultant? (Find the amplitude and polarization of the resultant wave).

Solution. It can be shown that the amplitude of the resultant wave is equal initial wave. To do this, use:

$$\begin{aligned}\mathbf{E}_{\text{res}} &= \mathbf{E}_1 + \mathbf{E}_2 \\ &= E_0 \hat{x} \cos(\omega t - kx) + E_0 \hat{y} \cos(\omega t - kx) \\ &= E_0 (\hat{x} + \hat{y}) \cos(\omega t - kx)\end{aligned}$$

Therefore, the amplitude of the resultant wave is $\sqrt{2}E_0$, obtained using $|\mathbf{E}_{\text{res}}| = (\mathbf{E}_{\text{res}} \cdot \mathbf{E}_{\text{res}})^{1/2}$ and the polarization direction is $(\hat{x} + \hat{y})$, *i.e.* $\pi/4$ away from the x -axis.

- (c) In the above, consider the phase-difference between the two waves to be fixed at $\phi = \pi$. How does this affect the resultant wave?

Solution. Due to the phase π , one of the wave will pick up a $-$ sign.

- (d) Show that the the amplitude of the wave obtained by superposition of two waves of equal amplitude and a phase difference of $\phi = \pi/2$ is equal to the amplitude of the initial waves. What does the wavefront of the superposed wave look like?

What happens for $\phi = -\pi/2$?

Comment: The phase difference of $\pi/2$ leads to circularly polarized light. Depending on the sign of the phase difference, one obtains left or right circularly polarized light. The nomenclature for left/right is somewhat subjective in literature. However, an accepted practice is to look towards the direction of propagation.

Solution. Again, the resultant wave is given as:

$$\begin{aligned}\mathbf{E}_{\text{res}} &= \mathbf{E}_1 + \mathbf{E}_2 \\ &= E_0 \hat{x} \cos(\omega t - kx) + E_0 \hat{y} \sin(\omega t - kx)\end{aligned}$$

Therefore, the amplitude of the resultant wave is obtained as:

$$\begin{aligned}E_{\text{res}} &= (\mathbf{E}_{\text{res}} \cdot \mathbf{E}_{\text{res}})^{1/2} \\ &= E_0\end{aligned}$$

The situation does not change for $\phi = -\pi/2$.

8. The magnetic field equations for an electromagnetic wave in free space are

$$\begin{aligned}B_x &= B \sin(ky + \omega t) \\ B_y &= 0 \\ B_z &= 0.\end{aligned}$$

- (a) What is the direction of propagation? (b) Write the electric field equations. (c) Is the wave polarized? If so, in which direction.

Solution. The magnetic field is only along the x while the wave is propagating along the $-y$ direction. Therefore, the electric field which must be along a direction perpendicular to

both the propagating direction as well as the magnetic field, is along z . The equation for the electric fields is given by:

$$E_x = 0$$

$$E_y = 0$$

$$E_z = (-cB)\sin(ky + \omega t),$$

where the strength of the electric field is cB and $-$ sign comes from the condition of mutual perpendicularity ($\vec{k} \sim \vec{E} \times \vec{B}$).

Yes, the wave is polarized (along z).

9. Interference

Prove the cosine law: the path difference between the interfering beams in a (parallel) thin film geometry at oblique incidence is $2\mu t \cos \theta_r$, where μ is the refractive index of the denser medium, t is the thickness of the medium, and θ_r is the angle of refraction with respect to normal.

Solution. Consider the geometry shown below in Fig. 2. Path difference (p.d.) between the

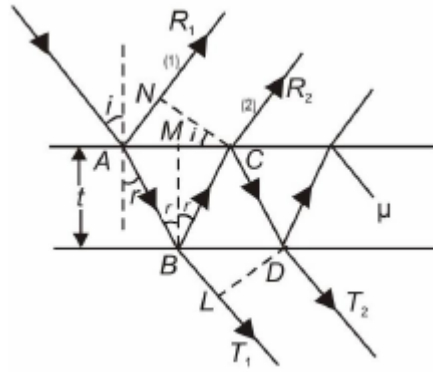


FIG. 2. Q6. The geometry of parallel thin films. N is the normal to R_1 from C , M is the perpendicular bisector to the top surface. Angles i and r are the angles of incidence, θ_i , and refraction, θ_r , respectively.

first reflected beams from the top surface (R_1) and the bottom surface R_2 is:

$$\text{p.d.} = \mu(AB + BC) - AN$$

$$\text{and, } AN = AC \sin \theta_i$$

Now, simple geometric considerations imply:

$$AB = BC = \frac{BM}{\cos \theta} = \frac{t}{\cos \theta}$$

Also,

$$\begin{aligned} AC = AM + MC &= BM \tan \theta_r + MN \tan \theta_r = 2t \tan \theta_r \\ \implies AN &= 2t \tan \theta_r \sin \theta_i \end{aligned}$$

Using Snell's law:

$$\begin{aligned} AN &= 2t \frac{\sin \theta_r}{\cos \theta_r} \mu \sin \theta_r \\ &= 2\mu t \frac{\sin^2 \theta_r}{\cos \theta_r} \end{aligned}$$

Putting AB, BC and AN together, we get:

$$\begin{aligned} \text{p.d.} &= \mu \left(\frac{t}{\cos \theta_r} + \frac{t}{\cos \theta_r} \right) - 2\mu t \frac{\sin^2 \theta_r}{\cos \theta_r} \\ &= 2\mu t \cos \theta_r \end{aligned}$$

10. [!] A thin glass of refractive index 1.5 is placed in the path and the central bright band is now shifted to the position of the third bright band. The fringes are produced by monochromatic light of wavelength 5450 Å. Calculate the thickness of the glass. You may assume any suitable geometry for interference.

Comment: (Hint) Compute the optical path difference due to (additional) glass in the path of the second interfering wave and equate it to twice/thrice the fringe width.

Solution. In order to solve this problem, one needs to assume an interference setup either in the form of a thin film wedge geometry (normal incidence), or Newton's rings, or the double slit interference. It is perhaps simplest to consider the case(s) where the fringe width is uniform, such as in wedge geometry or the double slit interference.

The optical path difference due to glass will be given by:

$$\text{OPD} = (\mu - 1)t + \lambda/2,$$

where the first term is the OPD due to distance travelled by the second wave in the medium of refractive index μ , $\lambda/2$ at the incident boundary due to phase shift when incident from a rarer medium to a denser medium. ($-t$ since the second wave would have travelled the distance t in the absence of glass anyways.)

The above OPD should be equated to 2β (or, 3β if the central bright band is counted as zeroth), where β is the fringe width to solve for t .

11. A thin film of thickness 4×10^{-5} cm is illuminated by white light normal to its surface. Its refractive index is 1.5. Of what color will the thin film appear in reflected light?

Solution. The color in the reflection will appear due to constructive interference.

The condition for constructive interference is:

$$2\mu t \cos(\theta) = (2n + 1)\frac{\lambda}{2},$$

where θ is the angle of incidence (w.r.t. to normal).

Plugging in $\theta = 0$, $\mu = 1.5$ and $t = 4 \times 10^{-5}$ cm, we get:

$$\lambda = \frac{2400}{2n + 1}(\text{nm})$$

For $n = 0, 1, 2, 3$, we get $\lambda = 2400, 800, 480, 343.1$ nm, respectively. As 2400 nm and, nm and 343.1 nm are not in the visible range, we will see only 480 nm (blue/bluish green) with naked eye.

12. White light is reflected normally from a uniform oil film ($n = 1.33$). An interference maximum for 6000 Å and a minimum for 4500 Å with no minimum in between, are observed. Calculate the thickness of the film.

Solution. The condition for interference maxima in the light reflected *normally* from the film of thickness t is:

$$2\mu t = \left(n + \frac{1}{2}\right)\lambda,$$

and that for a minima is:

$$2\mu t = n\lambda.$$

Using the value of $\mu = 1.33$ and respective wavelengths for maxima and minima, we end up with two algebraic equation with a single variable. **Note: One has to consider the $(n + 1)$ -th minima for the equations to be meaningful.**

Finally, one obtains $t = 3383 \text{ \AA}$.

13. [!] Consider a light incident on parallel plate(s) of refractive index $n_2 = 1.414$ ($n_1 = 1$ for air) at an angle on 45° . Obtain the possible values of wavelength which will produce bright fringes if the thickness of the parallel plates is t .
14. [!] Prove that two linearly polarized light waves of equal amplitude, with their planes of vibration at right angles to each other, cannot produce interference effects. (*Hint: Prove that the intensity of the resultant light wave, averaged over one or more cycles of oscillation, is the same no matter what phase difference exists between two waves.*)

Solution. Let the waves be travelling along $-z$ and polarized along the x and y directions, respectively. The net electric field is given by:

$$\begin{aligned} E^2 &= E_x^2 + E_y^2 \\ &= E_0(\sin^2(kz - \omega t) + \sin^2(kz - \omega t + \beta)) \end{aligned}$$

where β is the phase difference between the two waves.

We can consider any point, including $z = 0$ and average the result over a full cycle. Since β in the second term only shifts the integral limits, the results are independent of β . Therefore, no interference effects.

15. Interference from wedge-shaped surfaces

Light of wavelength $\lambda = 6300 \text{ \AA}$ falls normally on a thin wedge-shaped film ($n = 1.5$). There are ten bright and nine dark fringes over the length of the film. By how much does the film thickness change over this length?

Solution. The condition for destructive interference between light reflected from a film is:

$$2\mu t \cos \theta = n\lambda,$$

where θ is the incidence angle (w.r.t. normal).

Suppose the film thickness changes from t to $t + \delta t$ over the length. Let n be the order of the dark fringe appearing at one end of the film. The order of dark fringe at the other end will be $n + 9$. Therefore, we have

$$\begin{aligned} 2\mu t \cos \theta &= n\lambda \\ 2\mu(t + \delta t) \cos \theta &= (n + 9)\lambda \end{aligned}$$

Therefore, we get:

$$\begin{aligned} 2\mu\delta t \cos \theta &= 9\lambda \\ \Rightarrow t &= \frac{9\lambda}{2\mu \cos \theta} \end{aligned}$$

For normal incidence, $\cos \theta = 1$, leading to $t = 1.89 \times 10^{-4}$ cm.

16. Two glass plates 12 cm long touch at one end, and are separated by a wire 0.048 mm in diameter at the other. How many bright fringes will be observed over the 12 cm distance in the light ($\lambda = 6800 \text{ \AA}$) reflected normally from the plates? (Take $\mu = 1$).

Solution. Let t be the thickness of the wire and l is the length of the wedge, as shown in

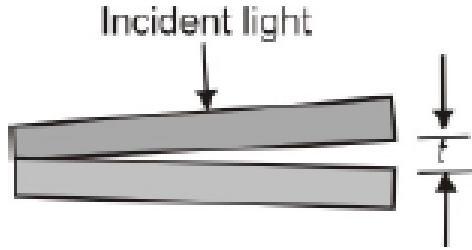


FIG. 3. Q10

the adjoining Fig. 3. The wedge angle is:

$$\theta = t/l \text{ (in radian)}$$

Now, the fringe width is $\beta = \lambda/2\theta$. Or,

$$\beta = \frac{l\lambda}{2t}$$

Since N fringes are seen, $l = N\beta$. Thus,

$$\beta = \frac{N\beta\lambda}{2t}$$

$$\therefore N = \frac{2t}{\lambda}$$

Plugging in the values of $\lambda = 6800 \text{ \AA}$, and $t = 0.048 \text{ nm}$ (and appropriately converting the units, we get $N = 141$).

17. Newton's rings

- (a) What is an efficient (and presumably error-free) way to obtain λ ?

Solution. The best way would be to compute the difference between $(m+p)$ and m -th ring:

$$\lambda = \frac{D_{m+p}^2 - D_m^2}{4pR} \quad (\text{typically } p \approx 10)$$

Here, D_m stands for the diameter of the m -th ring.

- (b) Show that in the Newton's rings arrangement, if the plano-convex lens is slowly lifted vertically upwards, the fringes will move inwards.

Solution. Example 15.2 in Ghatak's book, presumably 1st edition! (page attached at the end)

- (c) Newton's rings are formed in reflected light of wavelength $5895 \text{ \AA}$ with a liquid between the plane and curved surfaces. The diameter of the 5th ring is 0.3 cm and the radius of curvature of the curved surface is 100 cm . Calculate the refractive index of the liquid, when the 5th ring is i) bright, ii) dark.

Solution. The diameter D_n of the n -th bright ring is given by:

$$D_n^2 = \frac{2(2n_1)\lambda R}{\mu}$$

$$\therefore \mu = \frac{2(2n_1)\lambda R}{D_n^2}$$

Here, we have $n = 4$, $\lambda = 5895 \times 10^{-8} \text{ cm}$, $R = 100 \text{ cm}$, $D_n = 0.3 \text{ cm}$, leading to:

$$\mu = 1.18$$

Similarly, for the n -th dark ring, we have

$$D_n^2 = \frac{4n\lambda R}{\mu},$$

leading to $\mu = 1.31$.

| *Comment: Explicitly derive the conditions for observing a bright spot at the center.*

18. Solution to the transcendental equation $\tan(x) = x$.

Solution. Transcendental equation generally do not permit analytical solutions. One of the easiest way to obtain the solution is graphically. To this end, plot $\tan(x)$ and x and look at the points of intersection (see Fig. 4).

You will find: $x = 0$ is a trivial solution. Other solution(s) at $x = 1.43\pi, 2.46\pi, 3.47\pi, \dots$

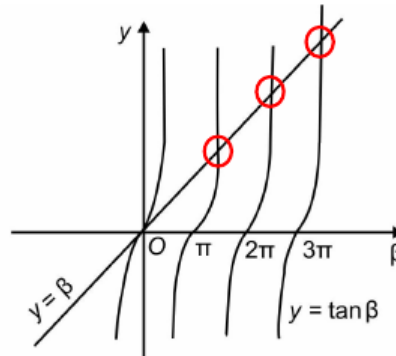


FIG. 4. Plot of $\tan(x)$ and x .

19. **Diffraction from a single slit.**

Sound wave has a speed of 330 m/s, and suppose it's frequency is 100 Hz. What would be the typical aperture size to experience appreciable diffraction?

Solution. $\lambda = v/f = 330/100 \text{ m} = 3.3 \text{ m}$.

Diffraction is observed if the aperture is at least of the size of λ .

20. **Diffraction from 2-slit arrangement.**

Explicitly derive the conditions for maxima/minima. Note and plot the intensity profile.

21. **Computational/Simulation exercise(s).**

Plot the resulting intensity patterns from (i) a single slit diffraction; (ii) double slit diffraction.

Solution. Check out these links:

- Single slit: [Geogebra](#) - based on the formula
- Young's double slit: [Geogebra](#), [Javalab](#) or, [oPhysics](#)

| *Comment: Study the dependence on the slit width, etc.*

22. **Light matter interaction.** To provide a physical intuition of how light interacts with matter to produce polarization and related effects, you may have heard about the fact that oscillating EM waves induce oscillations in electric dipoles. **Here** is a nice illustration/animation of the same even if the context is limited.

replace the red filter by a green filter in front of the eye, the fringe pattern corresponding to the green color will appear; this is similar to the discussion in Sec. 14.9.

Example 15.2 Consider the formation of Newton's rings by monochromatic light of $\lambda = 6.4 \times 10^{-5}$ cm. Assume the point of contact to be perfect. Now slowly raise the lens vertically above the plate. As the lens moves gradually away from the plate, discuss the ring pattern as seen through the microscope. Assume the radius of the convex surface to be 100 cm.

Solution: Since the point of contact is perfect, the central spot will be dark, the first dark ring will form at P where $PA = \lambda/2$, and the radius of this ring OA will be $\sqrt{\lambda R}$ ($= 0.080$ cm); see Fig. 15.32(a). Similarly, the radius of the second dark ring will be $OB = \sqrt{2\lambda R}$ ($= 0.113$ cm). If we now raise the lens by $\lambda/4$ ($= 1.6 \times 10^{-5}$ cm), then $2t$ corresponding to the central spot would be $\lambda/2$ and instead of the dark spot at the center we will now have a bright spot. The radii of the first and the second dark rings will be

$$OA_1 = \left(\frac{1}{2}\lambda R\right)^{1/2} = 0.0566 \text{ cm}$$

and
$$OB_1 = \left(\frac{3}{2}\lambda R\right)^{1/2} = 0.098 \text{ cm}$$

respectively [see Fig. 15.32(b)]. If the lens is further moved by $\lambda/4$ (see Fig. 15.32(c)), then the first dark ring collapses to the center and the central spot will be dark. The ring which was originally at Q now shifts to Q_2 ; similarly the ring at R [Fig. 15.32(a)] collapses to R_2 [Fig. 15.32(c)].

Thus, as the lens is moved upward, the rings collapse to the center. Hence if we can measure the distance by which the lens is moved upward and also count the number of dark spots that have collapsed to the center, we can determine the wavelength. For example, in the present case, if the lens is moved by 6.4×10^{-3} cm, 200 rings will collapse to the center. If one carries out this experiment, it will be observed that the 200th dark ring will slowly converge to the center, and when the lens has moved by exactly 6.4×10^{-3} cm, it has exactly come to the center.

Example 15.3 Consider the formation of Newton's rings when two closely spaced wavelengths are present; for example, the D_1 and D_2 lines of sodium ($\lambda_1 = 5890 \text{ \AA}$ and $\lambda_2 = 5896 \text{ \AA}$). What will be the effect of the presence of these two wavelengths as the lens is gradually moved away from the plate? What will happen if the sodium lamp is replaced by a white light source?

Solution: We will first assume that the lens is in contact with the plane glass plate [see Fig. 15.32(a)]. Since the two wavelengths are very close, the bright and dark rings of λ_1 superpose on the bright and dark rings of λ_2 , respectively. This can be easily seen by calculating the radii of the ninth dark and bright ring for each wavelength.

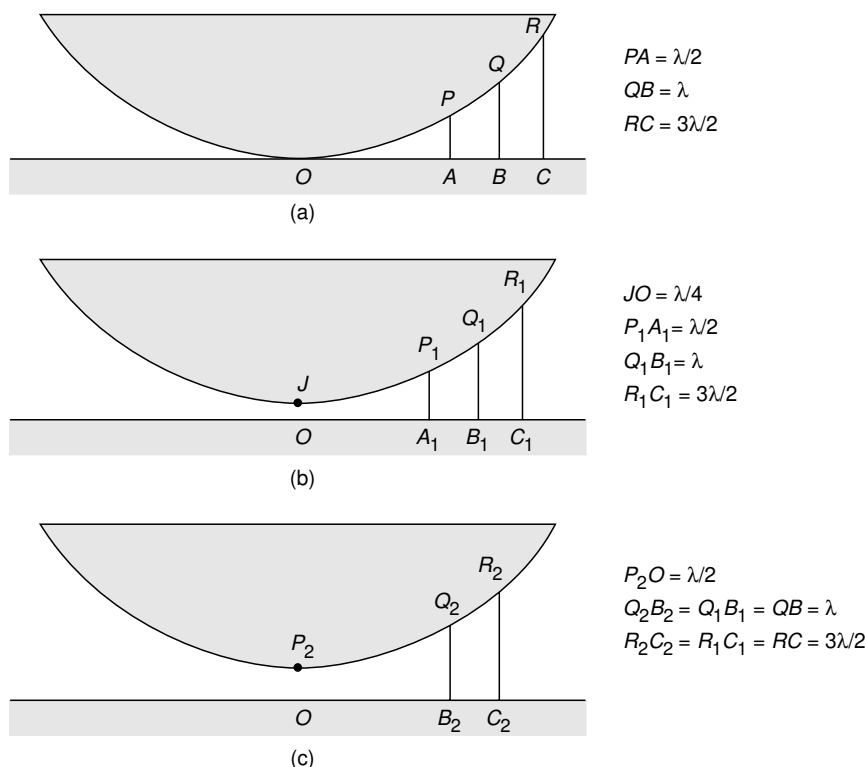


Fig. 15.32 The rings collapse to the center as the lens is moved away from the plate.